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Integrating Non-matrix Features into Permeability Modeling Using a Core-based 3D Probability Approach for Heterogeneous Carbonate Reservoirs, Offshore Abu Dhabi

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Abstract

Accurate permeability modeling is critical for reliable production forecasting in carbonate reservoirs, where conventional matrix-based models often significantly underestimate the actual flow capacity observed in well tests. This study introduces a novel approach that integrates non-matrix geological features—specifically dissolution vugs—into a three-dimensional, probability-based permeability model to improve the accuracy of dynamic reservoir simulations.

The methodology was applied to a complex offshore carbonate reservoir located in Abu Dhabi, United Arab Emirates. Detailed core analyses identified the spatial distribution of dissolution vugs and their correlation with key reservoir properties such as rock types and porosity. These non-matrix features were found to be concentrated in high-energy depositional environments, which were effectively captured using Static Rock Typing (SRT) as a proxy. A probabilistic model was then developed using SRT logs from over fifty-one wells across the field, enabling the enhancement of the permeability distribution by incorporating non-matrix elements.

This integrated approach provides a geologically consistent permeability model that better aligns with well test data, enhances the accuracy of history matching, and supports improved reservoir performance forecasting and optimization. It offers a robust framework for addressing lateral heterogeneity and improving fluid flow predictions in complex carbonate systems.

Introduction

Porosity and permeability variations often observed in carbonate rocks can be simply explained by their original depositional settings and the diagenetic alterations they have undergone over geological time. These diagenetic alterations such as dissolution, cementation and compaction modify the rock fabric and lead to highly complex pore systems. In many cases, these diagenetic changes are not uniform across the reservoir, providing zones of both enhanced and reduced permeability. This inherent complexity presents significant

challenges for reservoir engineers, particularly when trying to achieve accurate permeability predication in reservoir simulation studies (Yamamoto et al., 2025).

In conventional static modeling workflow, matrix permeability is typically derived from measurements obtained at the core-plug scale. These measurements are valuable for understanding small-scale rock properties; however, they often underestimate the actual flow capacity observed during well-tests. This discrepancy between predicated and measured permeability is primarily due to the influence of geological non-matrix features such as dissolution vugs and fractures, which mainly affect the lateral reservoir heterogeneity and act as high permeability that strongly influences the overall fluid flow behavior (Montani et al., 2025).

The present study addresses these discrepancies between matrix-derived and well-test permeability by incorporating geological non-matrix features – particularly dissolution vugs- into the model process. This integration results in a more geologically reasonable permeability model and enhances history matching performance. The approach is implemented through the development of a three-dimensional probability-based workflow, specifically designed for a heterogenous carbonate reservoir setting.

The study is conducted on a large offshore oilfield located northwest Abu Dhabi, United Arab Emirates (Figure 1). The field extends approximately 40 kilometers along the east to west direction and reaches a maximum width of about 25 kilometers. It was discovered in the early 1960's through seismic surveys. Structurally, the field is characterized as a low-amplitude anticline, gently dipping toward the east and northeast at an angle of around 2.5 degrees (Aldabal and Alsharhan, 1989; Alsharhan, 1990). In addition, interpretation of regional gravity anomalies indicates that the field structure has developed above the crest of underlying salt pillows, which have played a key role in shaping the reservoir architecture (Geng et al., 2020).

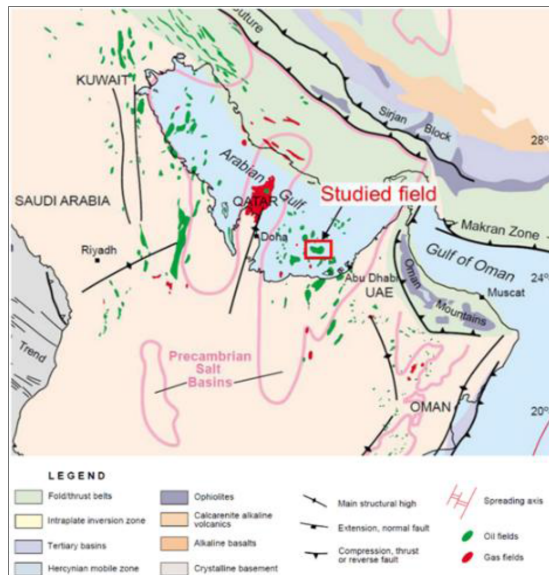


Figure 1—Location map of the oil and gas fields in the Arabian Gulf region (modified after Konert et al., 2001). The red square indicates the studied field, offshore Abu Dhabi.

Geological Setting

The studied reservoir, part of the Lower Cretaceous, was deposited in an intertidal zone within a moderate to low energy inner ramp setting (Figure 2). This vertical succession reflects a clear shallowing upward trend, transitioning from high-energy, grain-dominated deposits at the base to shallower, low-energy, mud-dominated deposits at the top of the regressive cycles (Kusuma et al.2024; Seksaf et al., 2024). Lithologically, the thick carbonate reservoir consists of both limestone and dolomite and is divided into

two main units based on variations in reservoir properties and depositional settings (Figure 3); a highly heterogeneous upper unit and a less heterogeneous lower unit. The upper unit mainly consists of mud-dominated limestone facies and microcrystalline dolomite, reflecting deposition in a low-energy inner ramp/intertidal setting. In contrast, the lower unit is dominated by grain-supported facies, mainly grainstones with dolomite being less abundant overall compared to the upper unit. This interval was deposited in a moderate-to high-energy inner ramp setting (Kusuma et al.2025). Both units show variations in facies laterally, which affect the reservoir’s heterogeneity, how different parts are connected, and how fluids can move through the rock. In addition, early diagenetic processes—such as cementation and dolomitization—occurred unevenly across the reservoir, further influencing rock quality and adding to the complexity in both vertical and horizontal directions.

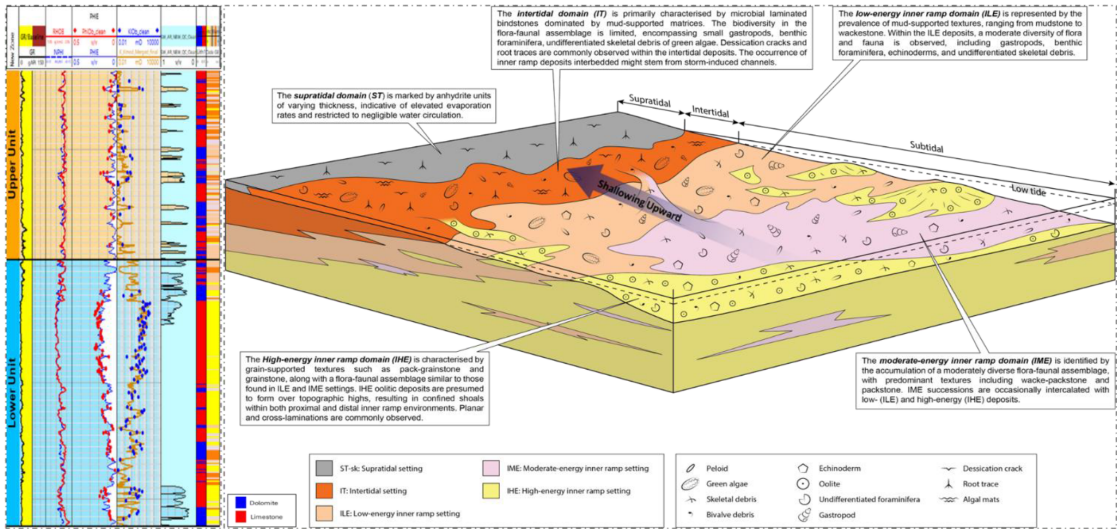


Figure 2—Conceptual depositional model of study area that show shallowing upward, from predominant poorly-moderately-cemented, high-energy, inner ramp (grainstone-dominated) deposits to predominant moderately/strongly cemented, mud-supported, intertidal deposits (Kusuma et al.2024; Seksaf et al., 2024).

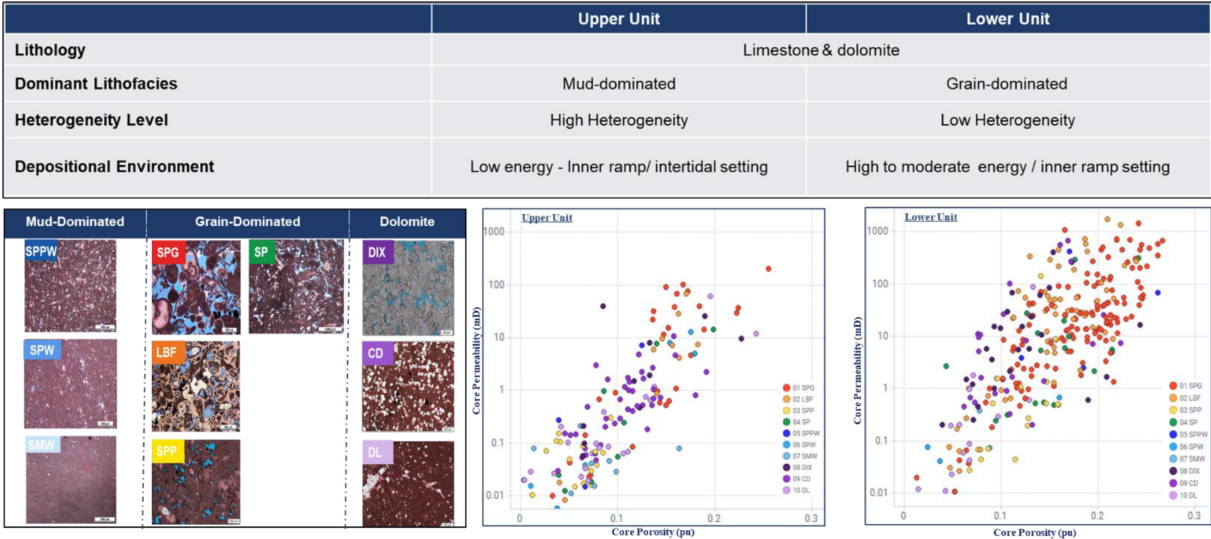


Figure 3—Summary of carbonate reservoir units showing lithological and depositional differences. Thin sections images illustrate dominant facies types, while Phi-K plots highlight heterogeneity contrast between the mud-supported upper unit and grain-supported lower unit (Kusuma et al.2024; Kusuma et al.2025).

Field Development History

The reservoir, discovered based on seismic data, began oil production in 2018. Since its discovery, seven wells have been cored. Including four that are fully cored, providing valuable information about the reservoir's rock properties. In addition, over fifty wells have been logged and are available with standard tools such as Gamma ray (GR), Neutron-porosity (NPHI), and Density logs. Among these thirteen wells covered all the reservoir zones, providing a comprehensive dataset for static reservoir characterization. However, dynamic data for the target zone of this study is not yet available. The field remains in the early stage of evolution, with ongoing plans of dynamic data acquisition such as permeability test. The example presented in this study is based on data from a different zone and is included only for illustrative purposes. It serves to validate the permeability matching purpose and does not reflect actual results for the current study zone.

Requirements of Excess Permeability in Modelling Process

In conventional static modeling phase, permeability models are constructed from permeability derived from core plugs and logs data, which only reflect matrix permeability. These models fail to capture larger-scale geological non-matrix features such as vugs and fractures. However, at the whole-core or field scale, non-matrix features can have a significant impact on reservoir performance. Consequently, integrating non-matrix features into permeability modeling is critical for achieving an accurate reservoir simulation model (Figure 4). This study highlights why it is important to include excess permeability in reservoir models to accurately simulate fluid flow. The conventional permeability models are usually based on core plug data. These data provide a detailed view of the rock at a small scale, but they do not capture larger geological features that affect how fluids flow. At large scale (whole-core), features like big vugs, and fractures – known as non-matrix permeability features – provide extra pathways for fluid to flow. This can make actual permeability higher than what matrix-based models suggest. Including these features as excess permeability helps the model match well-test data and better represents real reservoir behavior. Ignoring these geological features can cause difference between matrix permeability and well-test result. This discrepancy makes it harder to predict production accurately and can lead to poor reservoir management. By including non-matrix permeability, models better reflect fluid flow, which improves forecasting, decision-making, and overall reservoir optimization.

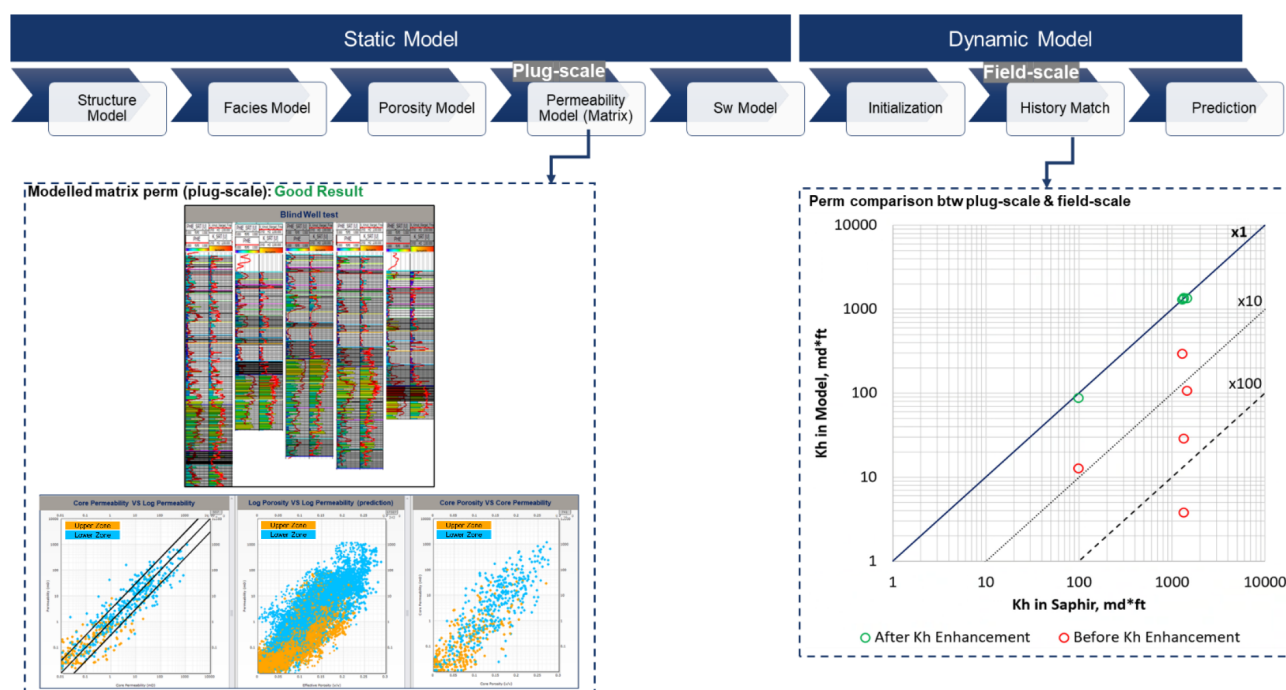


Figure 4—Plug-scale vs. Field-scale Permeability: The plug-scale matrix permeability model provides reasonable estimates; however, a clear discrepancy is observed when compared to field-scale permeability obtained from well test measurements.

Methodology

Static Rock Types Workflow

The Static Rock Typing (SRT) workflow followed a structured and integrated approach (Figure 5), designed to combine several datasets to ensure geological consistency and improve reservoir characterization. The study utilized data from five cored wells, which included Comprehensive Core Descriptions, Routine Core Analysis (RCA), and Mercury Injection Capillary Pressure (MICP) measurements. These datasets provided detailed information on lithology, porosity, permeability, and pore-throat distribution. In addition to core data, approximately fifty well logs were available with Gamma ray (GR), Neutron-porosity (NPHI), and Density logs. thirteen wells covered all reservoir zones, allowing the prediction of permeability and SRT logs throughout the field. The preliminary stage involved quality control of cores and logs data to ensure data reliability and consistency. Once the data were validated, a detailed geological analysis was conducted to define Candidate Rock Type (CRT) based on lithofacies, diagenesis, and micro-scale textural variations observed in the cores. In parallel, permeability prediction and Petrophysical Grouping (PG) classification were conducted by integrating RCA and MICP measurements with log-derived data. PG classification allowed quantitative grouping of rocks based on porosity, permeability, and pore-throat size, capturing the petrophysical variability that cannot be identified by visual core description alone. This was followed by the construction of Static Rock Types (SRT) by combining CRT and PG for each lithology, including limestone and dolomite. The resulting SRTs were then used for SRT log prediction, enabling the extension of core-derived knowledge to uncored wells. These SRT logs also served as constraints for subsequent property modeling (porosity and permeability) and saturation modeling (Sw), ensuring consistency across all reservoir simulation inputs. Consequently, this Static Rock Typing approach resulted in grouping Limestone into nine SRTs groups, while Dolomite was grouped into six SRT groups (Figure 6). This integrated approach ensured that both geological and petrophysical variability were captured, improving the reliability of the 3D static reservoir models used for field development planning and dynamic simulations.

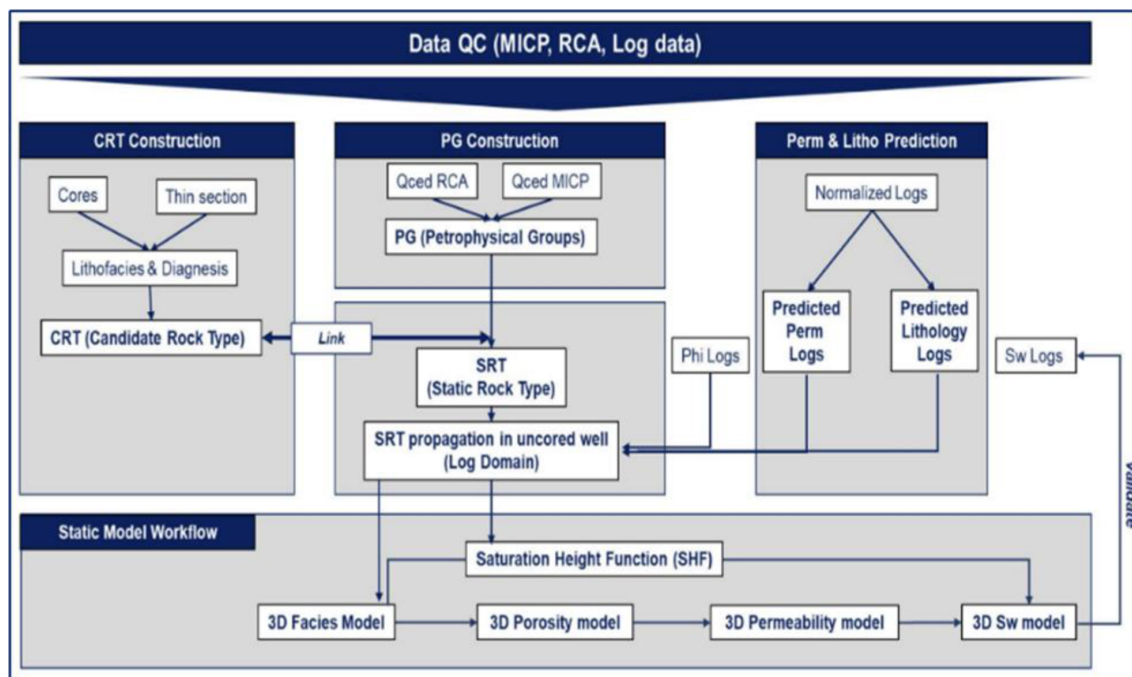


Figure 5—Static Rock Typing and Static Model Workflow (Montani et al., 2024).

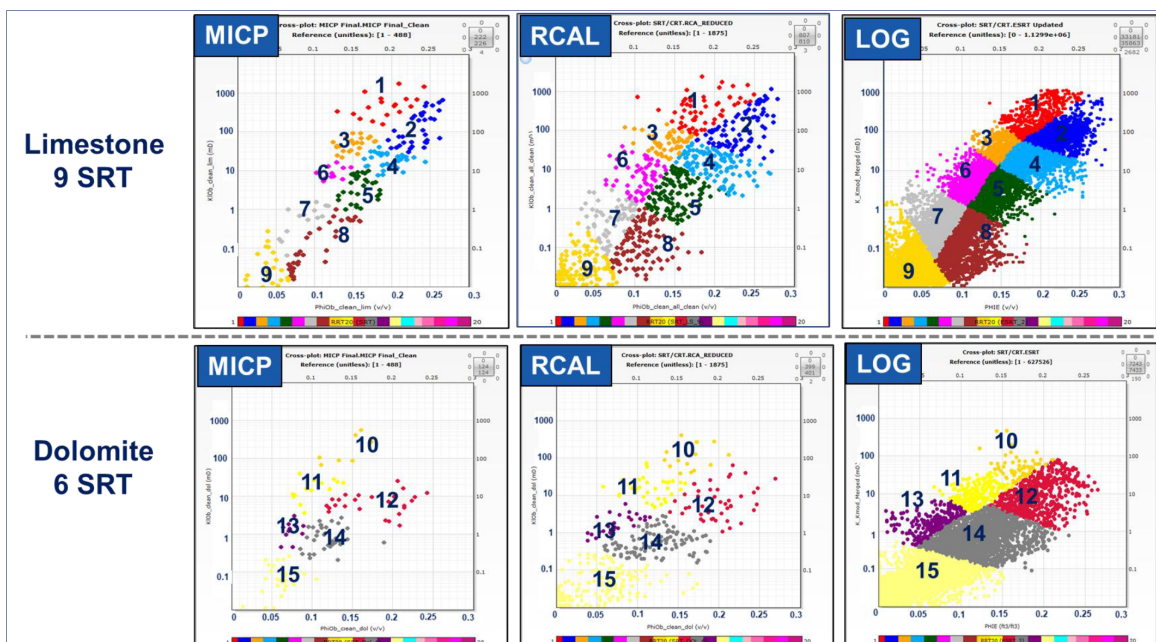


Figure 6—Final grouping of Static Rock Types based on PG-CRT classification in both Limestone and Dolomite (Kusuma et al.2024; Kusuma et al.2025), illustrating the distribution of petrophysical and textural characteristics across the studied units.

Dissolution Vug Concept (Non matrix description)

Since non-matrix features were not directly identified in the SRT logs, detailed core analyses were conducted to identify and flag dissolution vugs for each core plug (Figure 7). In total, more than 400 core plug samples were carefully examined, evaluated and correlated with key reservoir properties, including rock type, porosity, and permeability to better understand the influence of dissolution vugs on fluid flow. So, these observations identified a consistent presence of dissolution vugs that were localized within specific Static Rock Types (SRTs) that were most often associated with a high energy depositional environment.

Based on the observed distribution of vugs, a 3D probability model was developed and applied to the matrix permeability model, enhancing it by explicitly accounting for the effects of non-matrix features on reservoir flow. This approach allowed the reservoir model to better capture zones of enhanced fluid flow, minimizing differences between matrix-derived and well-test permeability. It also provides a more geologically realistic representation of connectivity and heterogeneity. By integrating both geological observations and probabilistic modeling, this methodology improves predictive accuracy in dynamic reservoir simulations, supports more reliable production forecasts, and offers valuable guidance for targeted reservoir management and decision-making.



Figure 7—Whole core and core plug samples showing the presence of dissolution vugs in studied carbonate reservoir.

Enhanced Permeability via Dissolution Vug Concept Model

Detailed core observations, combined with thin-section petrographic analysis, were used to flag dissolution vugs (and micro-fractures) in each core plug. This can confirm that these non-matrix features were concentrated within specific Static Rock Types (SRTs), demonstrating that SRT data could be used as a predictive proxy for non-matrix elements. Therefore, using over fifty-one SRT logs distributed across the field at spacings of approximately 1–4 km, a spatially dense probability model for dissolution vugs was created (Figure 8). The depositional environment of the dissolution vugs is also addressed here. It has shown that the SRT log has a significant correlation with depositional environments and is generally interpreted to have been deposited in a high-energy depositional environment. In other words, the presence of a large number of vugs within a specific SRT implies that vugs are predominantly found in high-energy depositional environments, a common characteristic in carbonate reservoirs (Figure 9).

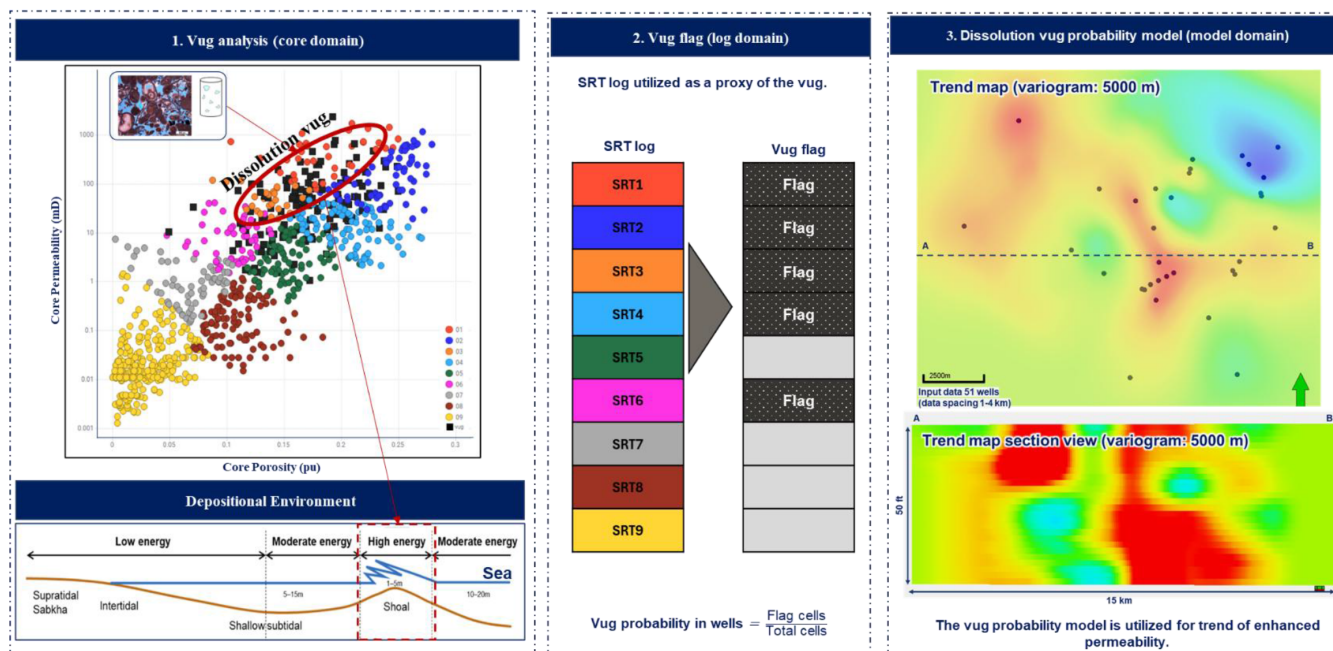


Figure 8—Dissolution vug probability model based on SRT logs (Display specific layer in dissolution vug probability model).

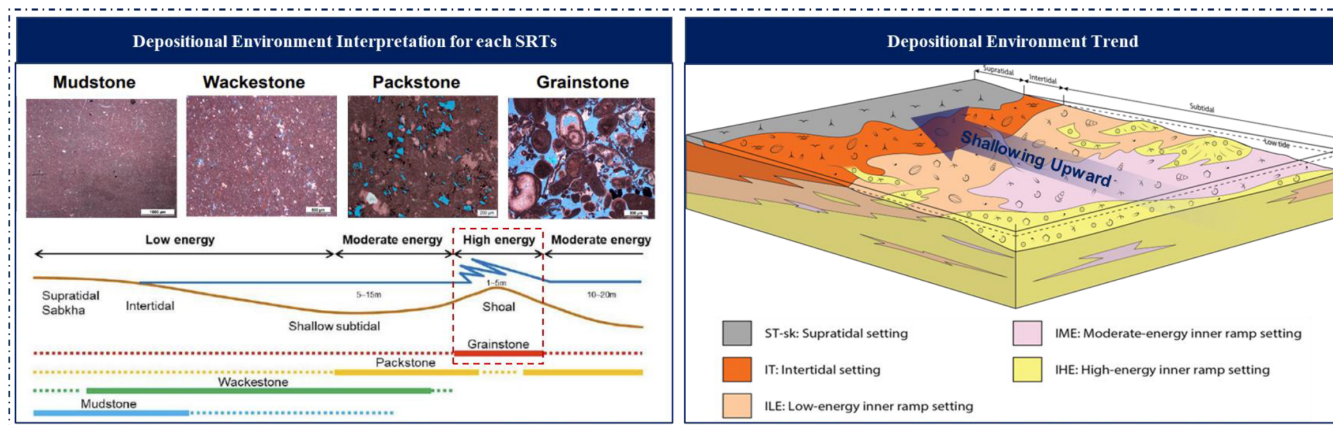


Figure 9—Depositional facies Dissolution vug deposition setting.

Optimized History Matching Using Cell Connectivity Adjustment in SRT Model

As previously described, no dynamic data is currently available for the target zone. In other words, due to the absence of well test-derived permeability, it is not yet possible to populate the dynamic model with specific permeability values. However, dynamic data acquisition is planned for this zone in the near future. Once well test permeability becomes available, it will be essential to reconcile it with matrix permeability.

To achieve this, a geological concept must be incorporated into the 3D model to guide how the matrix permeability should match the well-test data. In this study, the underlying geological concept is the spatial distribution of dissolution vugs, which, as mentioned earlier, is governed by the distribution of Static Rock Types (SRTs). Therefore, the dynamic model can be refined by modifying the SRT distribution accordingly. Additionally, connectivity is a key factor influencing dynamic model behavior. Various connectivity scenarios can be constructed by altering the variogram of the SRT log model (Figure 10). Among these scenarios, the most appropriate case can be selected, fine-tuned, and applied to the field model for further simulation and evaluation. The base-case variogram will be finalized during the history-matching phase,

once the well-test data are available then will selected the best match variogram value as the representative case.

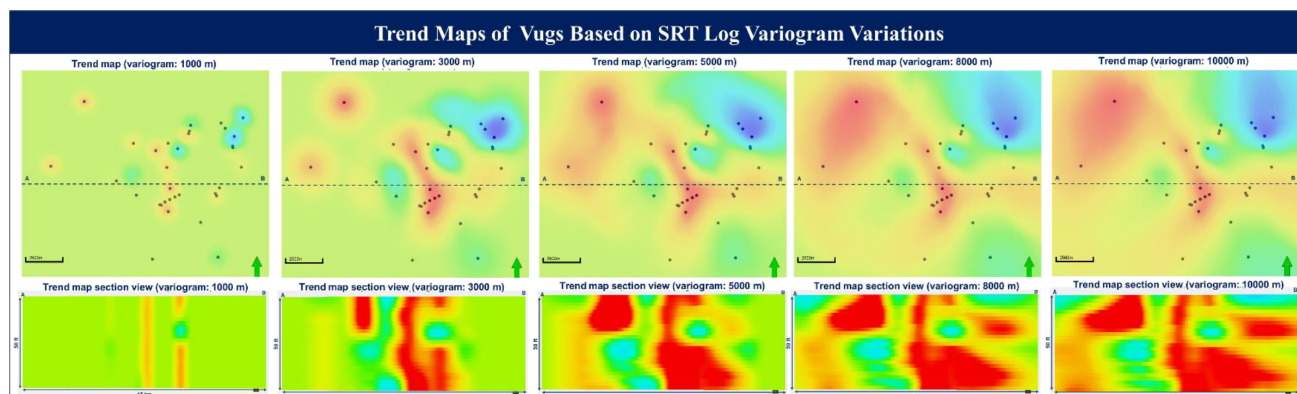


Figure 10—Examples of vug trend maps derived from SRT logs, illustrating different connectivity scenarios resulting from variogram variations.

Conclusion

This study presents a novel approach to permeability modeling in heterogeneous carbonate reservoirs by integrating non-matrix features—specifically dissolution vugs—into a 3D probability-based framework. Other non-matrix features like fractures are not confirmed from core or image log data in the target zone of his study. Traditional matrix-based permeability models often fail to capture the actual flow capacity observed in well tests, primarily due to the absence of non-matrix geological features. By leveraging detailed core analysis and static rock typing (SRT), this study successfully identified the spatial distribution of dissolution vugs and established a strong correlation between these features and high-energy depositional environments.

The integration of dissolution vug probability into the permeability model significantly enhanced the geological realism and predictive capability of the static model. Furthermore, the use of SRT as a proxy for non-matrix features enabled the development of a geologically consistent permeability distribution, which can be dynamically refined through cell connectivity adjustments once well test data becomes available. This approach also helps to better understand how fluid flow varies at different scales, enabling more informed decisions in reservoir management and optimization of production strategies.

This methodology not only improves the alignment between static and dynamic models but also provides a robust framework for future history matching and reservoir simulation efforts. The approach is particularly valuable for complex carbonate systems where non-matrix features play a critical role in fluid flow behavior and an impact on optimizing both numbers for producers and injectors wells.

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